Bayesian Estimation of DSGE Models

Posterior inference and a richer RBC economy

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What do the data tell us about uncertainty?

Last session

The likelihood ranked parameter values through the model's prediction errors. A numerical optimum did not by itself establish identification.

The economic question

How uncertain are the mechanisms and their effects? Combine the sample with explicit prior information, without mistaking a prior restriction for evidence in the data.

Today

Build and sample the posterior, interpret Dynare diagnostics, then derive a richer RBC model whose shocks and observables have distinct economic roles.

Plan

1. Bayesian updating
2. Posterior simulation
3. Dynare inference
4. Full RBC model and measurement

The likelihood ranks parameter values. Priors make additional information explicit.

How we use the session

Teaching block	Effective minutes
Posterior construction and MH	45
Dynare priors and diagnostics	35
Full RBC model and linearization	65
Measurement, handout work and synthesis	35
Teaching time	180
Separate break before the full model	15

Keep historical figures as illustrations. New posterior claims require diagnostics for the actual sample and run.

The likelihood remains the same object

$$\ell(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[n_y \log(2\pi) + \log|\Omega_t| + S_t'\Omega_t^{-1}S_t\right]$$

- S_t and Ω_t come from the state-space filter.
- θ is the estimated subset of structural parameters.
- Each evaluation still requires an admissible solution and consistent measurement equations.

The statistical model starts from equilibrium

$$E_t[f_\Theta(x_{t+1}, x_t, x_{t-1}, \varepsilon_t)] = 0 \quad \Longrightarrow \quad \hat{x}_t = F(\Theta)\hat{x}_{t-1} + G(\Theta)\varepsilon_t.$$

$$y_t = H\hat{x}_t + v_t, \qquad S_t = Y_t - E_{t-1}y_t.$$

- Estimate the subvector θ while other components of Θ remain calibrated.
- For each candidate, solve the model, build the measurement equation and run the filter.
- Adding a prior changes the parameter criterion, not the meaning of the data or the filter.

Bayes combines two sources of information

$$p(\theta \mid Y) = \frac{L(\theta \mid Y)p(\theta)}{p(Y)} \propto L(\theta \mid Y)p(\theta)$$

- $p(\theta)$: prior information before using this sample.
- $p(\theta \mid Y)$: posterior information conditional on model and data.
- $p(Y) = \int L(\theta \mid Y)p(\theta)\, d\theta$: marginal likelihood.
- The normalizing constant is unnecessary for the mode or MH ratios.

Choose the support before the prior mean

Parameter	Support used here	Prior family
Positive persistence	$0 < \rho_A < 1$	Beta
Positive curvature	$(0, \infty)$	Gamma
Shock standard deviation	$(0, \infty)$	Inverse gamma
Unrestricted coefficient	$\mathbb{R}$	Normal

- Stationarity alone allows $-1 < \rho_A < 1$. Positive persistence is an additional restriction.
- Report the prior mean and s.d., justify dispersion and check sensitivity.

The teaching example estimates one variance

$$\varepsilon_t \sim \mathcal{N}(0, Q), \qquad Q_{\text{true}} = 1, \qquad T = 200$$

- Use the same three-variable RBC policy rule and invert simulated consumption.
- Only Q is estimated. The matrices F, G, H remain fixed.
- This $Q = 1$ experiment illustrates inference, not a realistic shock calibration.
- Distinguish the variance Q from the standard deviation $\sqrt{Q}$.

Recover the variance likelihood

With F, G, H fixed and invertible HG, the recovered shocks are fixed data:

$$\ell(Q) = -\frac{T}{2}\log(2\pi Q) - \frac{1}{2Q}\sum_{t=1}^{T}\hat{\varepsilon}_t^2, \qquad Q > 0.$$

$$\ell'(Q) = -\frac{T}{2Q} + \frac{\sum_t \hat{\varepsilon}_t^2}{2Q^2} = 0 \quad \Longrightarrow \quad \hat{Q}_{\text{MLE}} = \frac{1}{T}\sum_t \hat{\varepsilon}_t^2.$$

- The inversion Jacobian is constant when only Q changes.
- This formula provides a benchmark for the numerical optimizer.
- Likelihood reparameterization uses $Q = \sigma^2$. A transformed prior also needs $p_\sigma(\sigma) = 2\sigma p_Q(\sigma^2)$.

A prior restricted to positive variance

$$p(Q) \propto \exp\left[-\frac{1}{2}\left(\frac{Q-1}{0.10}\right)^2\right]\mathbf{1}_{\{Q>0\}}$$

- The handout uses a normal kernel with mean parameter 1 and s.d. parameter 0.10.
- Restrict it to $Q > 0$. The truncation constant cancels in mode and MH calculations.
- This tight illustrative prior is not a default prior for every shock variance.

Optimize the posterior kernel

$$g(\theta) = \ell(\theta) + \log p(\theta), \qquad \hat{\theta}^{\text{mode}} = \arg\min_{\theta}[-g(\theta)]$$

- The mode is the location of highest posterior density.
- The posterior mean averages parameter values over that density.
- Neither summary is generally a simple average of the prior mean and the MLE.

See how the prior changes the optimum

For the truncated normal kernel with $\mu_0 = 1$ and $s_0 = 0.1$,

$$g(Q) = \ell(Q) - \frac{(Q - \mu_0)^2}{2s_0^2} + \text{constant}.$$

$$g'(Q) = -\frac{T}{2Q} + \frac{\sum_t \hat{\varepsilon}_t^2}{2Q^2} - \frac{Q - \mu_0}{s_0^2} = 0.$$

- The data score and prior score balance at an interior posterior mode.
- Below μ_0, the prior score is positive. Above it, the score is negative.
- There is no general formula averaging the MLE and prior mean.

Curvature supplies a local proposal scale

$$\Sigma_\theta \simeq \left[-\nabla^2 g(\hat{\theta}^{\text{mode}})\right]^{-1}$$

- A sharper posterior peak implies a smaller local covariance.
- This is a Laplace approximation, not the full posterior.
- Check that the negative Hessian is positive definite before using it.
- Multiple modes or strong asymmetry can make a local approximation misleading.

MATLAB: separate the two objective functions

```
lnprior = @(Q) -0.5*((Q-1)/0.1).^2 ...
               -log(0.1*sqrt(2*pi));
theta_MLE = fmincon(@(Q) -llk(Q),theta0,[],[],[],[],1e-8);
theta_BE = fmincon(@(Q) -(llk(Q)+lnprior(Q)), ...
                   theta0,[],[],[],[],1e-8);
```

- Here `llk` returns the log-likelihood, unlike the session 3 minimization helper.
- The current script computes a finite-difference posterior Hessian.

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A mode locates a peak. Sampling describes uncertainty around and beyond that peak.

MH step 1: propose from the current state

$$\theta_i^{\text{cand}} = \theta_{i-1} + cLz_i, \qquad z_i \sim \mathcal{N}(0, I), \qquad LL' = \Sigma_\theta$$

- θ_{i-1} is the last retained draw, not always the mode.
- c scales the proposal standard deviation.
- The scalar script uses $P = \sqrt{\Sigma_\theta}$ and $c = 2.2$.

MH step 2: compare posterior densities

$$\alpha_i = \min\left\{1, \exp\left[g(\theta_i^{\text{cand}}) - g(\theta_{i-1})\right]\right\}$$

- For this symmetric random-walk proposal, proposal densities cancel.
- A higher-density candidate is accepted with probability 1.
- A lower-density candidate may still be accepted. We sample, rather than optimize.
- A candidate outside the support has log posterior $-\infty$.

MH step 3: retain a state even after rejection

$$\theta_i = \begin{cases} \theta_i^{\text{cand}} & u_i < \alpha_i \\ \theta_{i-1} & u_i \geq \alpha_i \end{cases}, \qquad u_i \sim \mathcal{U}(0,1)$$

- A rejection repeats the previous state in the chain.
- Discarding repeated states would distort posterior frequencies.
- Reject unsupported proposals. Do not clip them onto a boundary.

Mini-exercise: two MH decisions

$$g(\theta_{i-1}) = -10, \qquad g(\theta_i^{\text{cand}}) = -11, \qquad u_i = 0.20$$

- Compute the acceptance probability and decide whether to move.
- Repeat with $u_i = 0.50$. What is stored in the chain?
- Would an optimizer take the same step?

Oral correction: a downhill move can be useful

$$\alpha_i = e^{-1} \simeq 0.368$$

- With $u_i = 0.20$, accept the candidate even though its density is lower.
- With $u_i = 0.50$, reject and repeat the previous state.
- These moves allow the chain to explore the whole posterior.
- Keeping only improvements would concentrate at a mode instead of sampling uncertainty.

Burn-in is not a convergence certificate

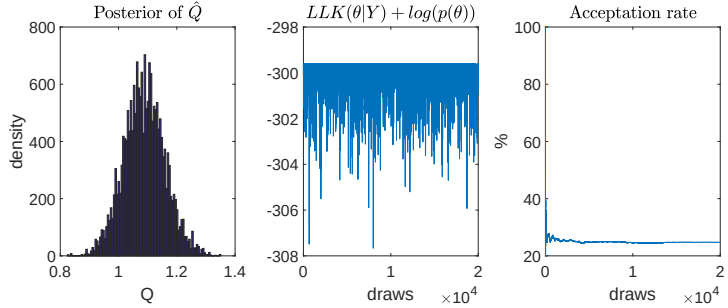
$$\widehat{E}[h(\theta) \mid Y] = \frac{1}{N - B} \sum_{i=B+1}^{N} h(\theta_i)$$

- B is the number of discarded initial draws, N is the full chain length.
- After burn-in, remaining autocorrelation reduces effective sample size.
- Compare dispersed chains, traces, autocorrelation and Monte Carlo stability.
- The supplied scalar example drops 10% but does not itself prove convergence.

Tune the proposal by looking at movement

- If c is too large, many proposals are rejected and the chain sticks.
- If c is too small, acceptance is high but exploration is slow.
- About 20–35% is a multivariate tuning heuristic, not a universal target.
- A plausible acceptance fraction cannot rule out nonconvergence or missing modes.

Read the three historical MCMC panels



- Left: shape of retained variance draws. Middle: log-posterior kernel by iteration.
- Right: cumulative acceptance, not a convergence certificate.
- The historical left panel shows counts despite its density label. Do not read heights as normalized probabilities.

Original handout MATLAB box 1. No new run. The current script normalizes its histogram.

MATLAB exercise: inspect the retained chain

```
run('bayesianRBC.m')
B = floor(0.1*numel(theta_mcmc));
kept = theta_mcmc(B+1:end);
plot(kept)
mean(kept)
sqrt(mean(kept))
mean(sqrt(kept))
```

- Which quantities summarize the variance and the standard deviation?
- Why are the last two numbers generally different?

Transform each draw before averaging

$$E(\sqrt{Q} \mid Y) \;\leq\; \sqrt{E(Q \mid Y)}$$

- The chain contains draws of Q. Its average estimates the posterior mean variance.
- To estimate the posterior mean s.d., transform every draw before averaging.
- The inequality follows from concavity of the square root.
- Use the canonical teaching file `bayesianRBC.m` for this experiment.

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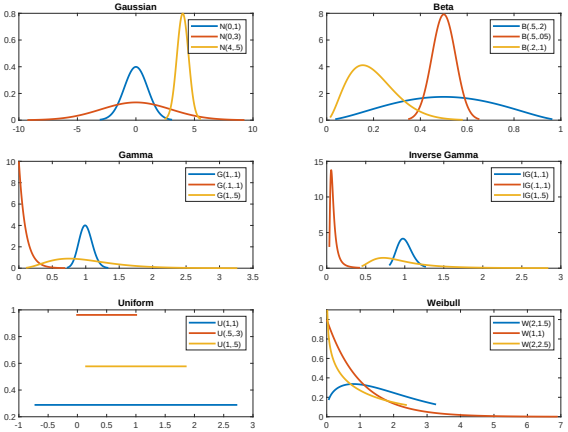
The scalar experiment clarifies the algorithm. Inspect priors, chains and IRFs in Dynare.

The prior block makes the assumptions explicit

```
estimated_params;
  stderr e_a, .7, 0, inf, inv_gamma_pdf, .1, 2;
  rho_A, .8, 0, 1, beta_pdf, .5, .2;
end;
varobs y;
```

- Read each line as initial value, lower bound, upper bound, shape, mean, s.d.
- ρ_A receives a beta prior. The shock s.d. receives an inverse-gamma prior.
- A prior can regularize a flat likelihood without creating data identification.

Prior shape and dispersion both matter



- Same support does not imply the same information.
- A tight prior restricts plausible values strongly.
- Dynare specifies means and standard deviations, not generally the native shape parameters.
- Uniform and Weibull families are also available.

Handout figure 2, historical illustration. Check economic support before selecting moments.

Shock scale determines the prior units

$$\hat{a}_t = \rho_A \hat{a}_{t-1} + \frac{e_{a,t}}{100}$$

- In `RBC_estim.mod`, y_t is a decimal log deviation.
- The s.d. of $e_{a,t}$ is therefore expressed in percentage points.
- An estimated s.d. of 1.1 corresponds to 0.011 in the technology equation.
- The full-model file later uses unscaled technology innovations. Its prior scale differs.

Dynare: simulate and retain enough evidence

```
estimation(datafile=mydata, first_obs=1,
           mh_replic=20000, mh_nblocks=2, mh_jscale=1.7,
           prefilter=1, bayesian_irf) y c k a;
```

- Use two chains and inspect their traces, autocorrelation and agreement.
- `prefilter=1` removes a mean, not a trend.
- `mh_jscale` controls proposal scale. More draws alone do not ensure convergence.
- Retain data, versions, mode files, chains and diagnostic output.

Run the Bayesian workflow

From `session4/codes`, prepare the data before either model:

```
which call_dbnomics
mydb
load mydata.mat y dy T
assert(all(isfinite(y)) && all(isfinite(dy)))
dynare RBC_estim.mod
% After understanding the richer model:
% dynare fullRBC.mod
```

- The simple model observes detrended y. The full model starts from growth dy and a different innovation scale.
- Inspect mode, chain diagnostics, burn-in and effective sample size before reporting posterior summaries.

Separate optimization from posterior sampling

The historical optimizer moves from $g \simeq 540.6$ to $g \simeq 895.6$. It minimizes $-g$, so a final negative objective is not a failure.

Parameter	Prior mean	Posterior mode	Local s.d.
ρ_A	.500	.9842	.0085
$\mathrm{sd}(e_a)$, pp	.100	1.1003	.0454

- The mode and Hessian summarize the local peak.
- MCMC next describes the distribution, its mean and credible intervals.
- Two chains of 20,000 draws are 40,000 draws before discarding burn-in, not 40,000 independent posterior observations.

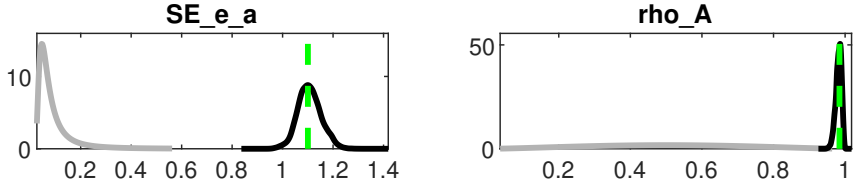
Handout section 2.2. Historical outputs, not current-sample estimates.

Read posterior summaries with their units

Historical quantity	Prior mean	Posterior mean	90% HPD interval
ρ_A	0.500	0.9817	[0.9690, 0.9946]
sd(e_a), pp	0.100	1.1064	[1.0335, 1.1844]

- Historical handout illustration, not a current-sample reproduction.
- An HPD interval describes posterior parameter uncertainty conditional on the model.
- The posterior mode, mean and local curvature s.d. answer different questions.

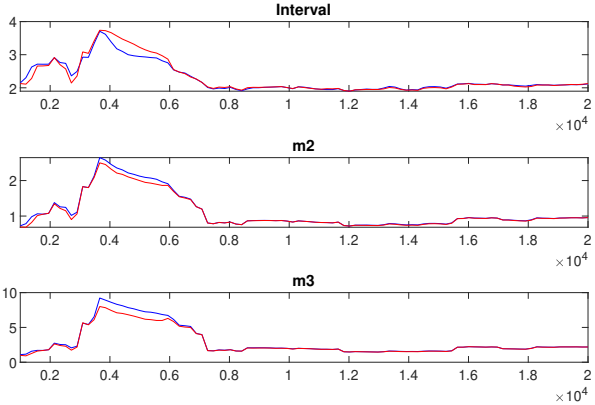
Learning can change the mean or the dispersion



Source: chapter 4, Estimation and posteriors. Historical illustration, not regenerated.

- Grey: prior. Black: posterior. Green: mean. Compare both width and location.

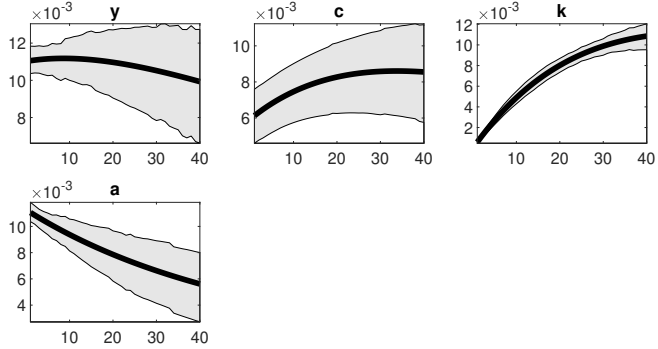
Compare the chains, not just one histogram



Source: chapter 4, Brooks–Gelman diagnostic. Historical illustration, not regenerated.

- Check whether within-chain and pooled measures become compatible. Inspect traces and mixing too.

Posterior IRFs propagate parameter uncertainty



- Bands come from posterior parameter draws for a conditional impulse.
- They do not describe future random shock sequences.
- This simple model has fixed labour. Endogenous hours enter only in the next model.

Handout section 2.3, historical illustration. No new simulation.

Reload an unchanged estimation

```
estimation(datafile=mydata, first_obs=1,
  mh_replic=0, prefilter=1, load_mh_file,
  mode_file=RBC_estim_mode, mode_compute=0) y c k a;
```

- Reuse only validated mode and chain files from the same experiment.
- Check the model, data, sample, transformations, parameter order and priors.
- Changing one of these can define a different posterior. Re-estimate it.

Read and work: improve the estimation

Read handout 4, §2.4, Exercise 1, *Fixing the estimation*, and work through all three questions.

- Checkpoint: acceptance is a tuning statistic. Which trace or mixing evidence supports a change in proposal scale?
- Before transferring a prior from Smets and Wouters (2007), check its parameter definition and moments.
- Explain why estimating α requires steady state and policy matrices to be recomputed.

Plan

1. Bayesian updating
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Inference needs economic mechanisms that can explain the chosen observables.

A richer economy separates four mechanisms

Block	New margin or disturbance
Household	Consumption, hours, bonds, investment and capital
Firms	Capital and labour demand under perfect competition
Government	Lump-sum-financed public spending
Shocks	Technology, spending, labour disutility, investment efficiency

- Start from the handout's nonlinear economy, then derive its linear approximation.
- The base model has 13 endogenous variables and 13 equations, before measurement equations.

Households trade consumption against hours

$$\max E_t \sum_{\tau \geq 0} \beta^\tau \left[\frac{C_{j,t+\tau}^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_{t+\tau}^H \chi \frac{H_{j,t+\tau}^{1+\sigma_L}}{1+\sigma_L} \right]$$

- C_{jt}: consumption, H_{jt}: hours, β: discount factor.
- σ_C and σ_L govern curvature. χ scales labour disutility.
- ε_t^H shifts labour disutility. The baseline has no habits.

The household budget

$$w_t H_{jt} + z_t K_{j,t-1} + r_{t-1} B_{j,t-1} = C_{jt} + I_{jt} + S(I_{jt}, I_{j,t-1}) + B_{jt} + T_{jt}$$

- w_t: wage, z_t: capital rental rate, r_t: gross bond return.
- B_{jt}: bonds, I_{jt}: investment, T_{jt}: lump-sum taxes.
- Adjustment costs S use goods. They are not inserted into the capital law.
- Suppress the household index from now on under symmetry.

Two constraints, two shadow values

Suppress j under symmetry. Define the budget surplus

$$b_t = w_t H_t + z_t K_{t-1} + r_{t-1} B_{t-1} - C_t - I_t - S(I_t, I_{t-1}) - B_t - T_t.$$

The household's Lagrangian is

$$\mathcal{L}_t = E_t \sum_{\tau \geq 0} \beta^\tau \Big[U(C_{t+\tau}, H_{t+\tau}) + \lambda^c_{t+\tau} b_{t+\tau} + \lambda^k_{t+\tau} \left(\varepsilon^I_{t+\tau} I_{t+\tau} + (1-\delta) K_{t+\tau-1} - K_{t+\tau} \right) \Big].$$

- λ^c_t: utility value of one more unit of available goods.
- λ^k_t: utility value of one more unit of installed capital.

Differentiate all five household choices

$$
\begin{aligned}
C_t: &\quad \lambda_t^c = C_t^{-\sigma_C}, \\
H_t: &\quad \varepsilon_t^H \chi H_t^{\sigma_L} = \lambda_t^c w_t, \\
B_t: &\quad \lambda_t^c = \beta r_t E_t \lambda_{t+1}^c, \\
I_t: &\quad -\lambda_t^c[1 + S_1(I_t, I_{t-1})] - \beta E_t[\lambda_{t+1}^c S_2(I_{t+1}, I_t)] + \varepsilon_t^I \lambda_t^k = 0, \\
K_t: &\quad \lambda_t^k = \beta E_t[(1-\delta)\lambda_{t+1}^k + \lambda_{t+1}^c z_{t+1}].
\end{aligned}
$$

Investment appears in two consecutive adjustment costs. Capital appears in today's accumulation and tomorrow's productive return. Keep both dates in each derivative.

Marginal utility prices the risk-free bond

$$\lambda_t^c = C_t^{-\sigma_C}, \qquad \lambda_t^c = \beta r_t E_t \lambda_{t+1}^c$$

$$1 = \beta r_t E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma_C} \right]$$

- λ_t^c is the multiplier on the budget constraint.
- One unit of consumption today is traded against tomorrow's marginal utility.
- The return is known when the bond is purchased.

Labour supply: marginal cost and benefit

$$\varepsilon_t^H \chi H_t^{\sigma_L} = \lambda_t^c w_t \quad \Rightarrow \quad w_t = \varepsilon_t^H \chi H_t^{\sigma_L} C_t^{\sigma_C}$$

- Working more is costly in utility but finances additional consumption.
- A higher labour-disutility shock raises the wage required for given C_t, H_t.
- In equilibrium, hours also depend on labour demand and the income effect.

Investment growth is costly to adjust

$$S(I_t, I_{t-1}) = \frac{\kappa}{2}\left(\frac{I_t}{I_{t-1}} - 1\right)^2 I_{t-1}$$

- $\kappa \geq 0$ governs adjustment costs.
- The cost depends on investment growth, not directly on the capital stock.
- At a constant-investment steady state, $S = 0$.
- Investment today affects costs today and costs tomorrow.

Differentiate the adjustment cost

Write the same function in a form that makes both arguments visible:

$$S(I, J) = \frac{\kappa}{2}\left(\frac{I^2}{J} - 2I + J\right).$$

$$S_1(I, J) = \frac{\kappa}{2}(2I/J - 2) = \kappa(I/J - 1),$$
$$S_2(I, J) = \frac{\kappa}{2}[1 - I^2/J^2].$$

- S_1: current marginal cost. S_2: effect on next period's cost base.
- Both derivatives vanish at $I = J = \bar{I}$.
- S_2 is a first partial derivative with respect to the second argument, not a second-order derivative.

Tobin's q links investment and value

$$q_t = \frac{\lambda_t^k}{\lambda_t^c}, \qquad \varepsilon_t^I q_t = 1 + S_1(I_t, I_{t-1}) + \beta E_t \left[\frac{\lambda_{t+1}^c}{\lambda_t^c} S_2(I_{t+1}, I_t) \right]$$

- q_t: shadow value of installed capital in consumption units.
- ε_t^I: efficiency with which investment creates capital.
- When $\kappa = 0$ and $\varepsilon_t^I = 1$, the condition gives $q_t = 1$.

Expand the investment condition

Use the two derivatives and $\lambda_t^c = C_t^{-\sigma_C}$:

$$\varepsilon_t^I q_t = 1 + \kappa \left(\frac{I_t}{I_{t-1}} - 1 \right) - \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma_C} \frac{\kappa}{2} \left(\left(\frac{I_{t+1}}{I_t} \right)^2 - 1 \right) \right].$$

- Current investment is costly when it rises relative to yesterday.
- It also raises the base for tomorrow, reducing the future adjustment cost.
- The future benefit is weighted by marginal utility and discounting.

Risky capital is priced by marginal utility

$$q_t = \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma_C} ((1-\delta) q_{t+1} + z_{t+1}) \right]$$

- The payoff combines the remaining capital value and the next rental payment.
- The stochastic discount factor weights payoffs by marginal utility.
- Replacing it by $1/r_t$ gives a first-order certainty-equivalent expression, not the exact risky-asset equation.

Investment efficiency and capital

$$K_t = \varepsilon_t^I I_t + (1-\delta)K_{t-1}$$
$$\hat{k}_t = \delta(\hat{\varepsilon}_t^I + \hat{\imath}_t) + (1-\delta)\hat{k}_{t-1}$$

- Capital available for production today is K_{t-1}.
- At steady state $\bar{I} = \delta\bar{K}$, which gives the weight δ.
- The investment-efficiency shock and investment share this same weight.

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Inference needs economic mechanisms that can explain the chosen observables.

Competitive firms pay marginal products

$$Y_t = \varepsilon_t^A K_{t-1}^{\alpha} H_t^{1-\alpha}$$

$$z_t = \alpha \frac{Y_t}{K_{t-1}}, \qquad w_t = (1-\alpha)\frac{Y_t}{H_t}$$

- ε_t^A: productivity, α: capital share.
- Current output can respond through productivity and hours, not current investment.
- Under constant returns and competition, factor payments exhaust output.

Why factor prices equal marginal products

Under competition, a firm's constrained profit objective is

$$\max_{Y,K_{-1},H} \; Y - zK_{-1} - wH + \lambda\big(\varepsilon^A K_{-1}^{\alpha} H^{1-\alpha} - Y\big).$$

$$1 - \lambda = 0, \qquad z = \lambda\alpha\frac{Y}{K_{-1}}, \qquad w = \lambda(1-\alpha)\frac{Y}{H}.$$

- Output is the numeraire, so marginal revenue is one and $\lambda = 1$.
- Equivalently, minimize $zK_{-1} + wH$ subject to producing a given Y.
- The multiplier is then marginal cost. Competition makes it equal to one.

Aggregation accounts for every use of output

$$Y_t = C_t + I_t + S(I_t, I_{t-1}) + g^y \bar{Y} \varepsilon_t^G$$

- Household and firm quantities coincide through labour and capital market clearing.
- Government spending is financed by lump-sum taxes. g^y is its steady-state output share.
- At first order, S disappears from resources because its level and first derivatives vanish at steady state.

Four shocks, four economic channels

$$\log \varepsilon_t^j = \rho_j \log \varepsilon_{t-1}^j + \eta_t^j, \qquad \eta_t^j \sim \mathcal{N}(0, \sigma_j^2), \quad j \in \{A, G, H, I\}$$

- A: productivity. G: public demand.
- H: labour disutility. I: investment efficiency.
- Each shock level is normalized to one in steady state.
- The supplied baseline estimates only technology. The other innovation variances are zero.

Collect the four optimality restrictions

$$
\begin{aligned}
1 &= \beta r_t E_t[(C_{t+1}/C_t)^{-\sigma_C}], \\
w_t &= \varepsilon_t^H \chi H_t^{\sigma_L} C_t^{\sigma_C}, \\
q_t &= \beta E_t[(C_{t+1}/C_t)^{-\sigma_C}((1-\delta)q_{t+1} + z_{t+1})], \\
\varepsilon_t^I q_t &= 1 + \kappa(I_t/I_{t-1} - 1) \\
&\quad - \beta E_t[(C_{t+1}/C_t)^{-\sigma_C} \kappa((I_{t+1}/I_t)^2 - 1)/2].
\end{aligned}
$$

These four restrictions determine intertemporal saving, labour supply, capital valuation and investment adjustment jointly.

Close the thirteen-equation model

$$
\begin{aligned}
K_t &= \varepsilon_t^I I_t + (1-\delta)K_{t-1}, \\
Y_t &= \varepsilon_t^A K_{t-1}^{\alpha} H_t^{1-\alpha}, \\
z_t &= \alpha Y_t / K_{t-1}, \qquad w_t = (1-\alpha) Y_t / H_t, \\
Y_t &= C_t + I_t + S(I_t, I_{t-1}) + g^y \bar{Y} \varepsilon_t^G.
\end{aligned}
$$

Add the four AR(1) processes already specified. Together with the four optimality restrictions, these are 13 equations for

$$\{r_t, C_t, H_t, Y_t, K_t, I_t, q_t, z_t, w_t, \varepsilon_t^A, \varepsilon_t^G, \varepsilon_t^H, \varepsilon_t^I\}.$$

Market clearing equates household and firm demands for labour and capital. Measurement equations are added only after this economic system is closed.

Steady state and calibration

$$\bar{r} = \beta^{-1}, \quad \bar{q} = 1, \quad \bar{z} = \beta^{-1} - (1-\delta), \quad \bar{K} = \bar{H}(\bar{z}/\alpha)^{1/(\alpha-1)}$$
$$\bar{Y} = \bar{K}^{\alpha}\bar{H}^{1-\alpha}, \quad \bar{I} = \delta\bar{K}, \quad \bar{C} = (1-g^{y})\bar{Y} - \bar{I}$$

- Baseline code: $\beta = .99, \delta = .025, \alpha = .36, g^{y} = .2, \bar{H} = 1/3$.
- Choose $\chi = \bar{w}/(\bar{H}^{\sigma_L}\bar{C}^{\sigma_C})$ to support the hours normalization.
- `fullRBC.mod` uses model-local coefficients: they update when their structural determinants change.

Linear household conditions

$$\frac{\hat{r}_t}{\sigma_C} = E_t\hat{c}_{t+1} - \hat{c}_t, \qquad \hat{w}_t = \hat{\varepsilon}_t^H + \sigma_L\hat{h}_t + \sigma_C\hat{c}_t$$

$$\hat{q}_t = \beta E_t\left[(1-\delta)\hat{q}_{t+1} + \bar{z}\hat{z}_{t+1}\right] - \hat{r}_t$$

- Hats denote log deviations from the deterministic steady state.
- The bond Euler equation substitutes expected consumption growth into capital pricing.
- These are the handout's baseline equations with no habits.

Linear investment keeps both growth terms

$$\hat{\varepsilon}_t^I + \hat{q}_t = \kappa(\hat{\imath}_t - \hat{\imath}_{t-1}) - \beta\kappa E_t(\hat{\imath}_{t+1} - \hat{\imath}_t)$$

- Current investment growth raises the current marginal adjustment cost.
- Expected future investment growth changes the value of adjusting today.
- No steady-state investment factor multiplies the first term for the specified S.
- This equation must be derived from the actual cost function, not imported from another RBC model.

Linear production, factor prices and resources

$$
\begin{aligned}
\hat{y}_t &= \hat{\varepsilon}^A_t + \alpha \hat{k}_{t-1} + (1-\alpha)\hat{h}_t, \\
\hat{z}_t &= \hat{y}_t - \hat{k}_{t-1}, \qquad \hat{w}_t = \hat{y}_t - \hat{h}_t, \\
\hat{y}_t &= \frac{\bar{C}}{\bar{Y}}\hat{c}_t + \frac{\bar{I}}{\bar{Y}}\hat{\imath}_t + g^y \hat{\varepsilon}^G_t, \\
\hat{\varepsilon}^j_t &= \rho_j \hat{\varepsilon}^j_{t-1} + \eta^j_t, \qquad j \in \{A, G, H, I\}.
\end{aligned}
$$

- $S(\bar{I}, \bar{I}) = S_1(\bar{I}, \bar{I}) = S_2(\bar{I}, \bar{I}) = 0$: no first-order resource cost.
- Its marginal conditions still affect investment because the next derivatives do not vanish.
- Combine these with the household and accumulation equations to recover all 13 linear restrictions.

Growth observables require their own equations

$$dy_t = 100(\hat{y}_t - \hat{y}_{t-1})$$

- The supplied series are percentage-point log changes. The data script rejects internal quarterly gaps before differencing.
- Raw growth has a mean. Remove it consistently or include a trend-growth intercept.
- The data script uses real quantities per labour-force participant, not per total resident.
- Hours use manufacturing weekly hours as an illustrative proxy, not aggregate hours.

Dynare exercise: add investment as an observable

```
// Add di to the endogenous declaration and model
di = 100*(i-i(-1));
// Then observe both growth series
varobs dy di;
```

- Activate the investment-efficiency innovation and estimate its s.d. and persistence.
- At the stationary steady state, $di = 0$. Check aligned data and covariance rank.
- Before running: verify the capital and adjustment-cost equations against the handout.

Build observables and shocks together

Observable	Measurement equation	Added innovation
Output growth	$dy_t = 100(\hat{y}_t - \hat{y}_{t-1})$	Technology
Investment growth	$di_t = 100(\hat{\imath}_t - \hat{\imath}_{t-1})$	Investment efficiency
Consumption growth	$dc_t = 100(\hat{c}_t - \hat{c}_{t-1})$	Public spending
Demeaned hours	$dh_t \simeq 100\bar{H}\hat{h}_t$	Labour disutility

- Add the variable, equation, zero steady state, aligned data and `varobs` entry.
- Estimate the corresponding shock persistence and standard deviation.
- At least as many independent innovations as exact observables is necessary, not sufficient: check forecast-error covariance rank.

From observables to decompositions

- Add $dc_t = 100(\hat{c}_t - \hat{c}_{t-1})$ with a spending shock.
- For demeaned hours in percentage points, $dh_t \simeq 100\bar{H}\hat{h}_t$. Activate the labour shock.
- Historical decomposition attributes the realized path to shock histories.
- Forecast-error variance decomposition attributes uncertainty at a chosen horizon. Neither equals the spending share g^y.
- Read and work handout 4, §3.6, Exercise 2, *Estimating the RBC model*. Follow its six stages and document the parameterization of imported priors.

Extension: external habits change marginal utility

$$\lambda_t^c = (C_t - hC_{t-1})^{-\sigma_C}, \qquad \hat{\lambda}_t^c = -\frac{\sigma_C}{1-h}(\hat{c}_t - h\hat{c}_{t-1})$$

$$\hat{r}_t = \frac{\sigma_C}{1-h}\left[E_t\hat{c}_{t+1} - (1+h)\hat{c}_t + h\hat{c}_{t-1}\right]$$

- The household treats lagged aggregate consumption as external. Read and work handout 4, Exercise 3, *External habits*, after deriving the baseline.
- Insert this marginal utility into labour supply, bond pricing and capital pricing.
- At steady state, $\bar{\lambda}^c = [(1-h)\bar{C}]^{-\sigma_C}$. To retain the same $\bar{H}$, recalibrate $\chi = \bar{w}\bar{\lambda}^c/\bar{H}^{\sigma_L}$.
- The code defaults to $h = 0$ and includes the $1-h$ normalization. Update all affected conditions together.

What makes posterior inference credible?

- Derive the economic model before estimating its linear system.
- Align priors, shock scales and measurement equations.
- Diagnose the mode, chains, effective information and prior sensitivity.
- Interpret posterior IRFs and decompositions conditional on those assumptions.
- Across the four sessions: measure fluctuations, solve the model, confront data, assess uncertainty.

Next step: use this chain to explain an estimate, its uncertainty and its economic interpretation.